Predicting Invariants of Finite p-Groups from Commuting Graphs

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This note summarizes the full paper (`paper.pdf`). It gives only the problem, the representation, and results that are proved or verified by computation. Empirical values are quoted from the experiments reported in the paper.

1 Problem and representation

We take every nontrivial 2-group of order at most 128 (2668 groups, via GAP's SmallGroups library) and build the *commuting graph* $\Gamma(G)$: vertices are group elements, edges are commuting pairs. Each vertex carries five features: element order, centralizer size, conjugacy class size, a centrality indicator, and *power depth*, the largest k with $x \in G^{2^k} = \langle g^{2^k} : g \in G \rangle$. The identity is assigned the terminal filtration index $e = \log_2 \exp(G)$, where $G^{2^e} = \{1\}$. A 3-layer Graph Isomorphism Network with mean pooling (optionally concatenated with a persistent-homology descriptor) regresses five targets: nilpotency class, derived length, $\log_2 |Z(G)|$, $\log_2 |[G, G]|$, and the number of conjugacy classes $k(G)$.

As well as measuring prediction error, we ask which targets the representation determines at all. If it does not determine a target, every model that reads it has some unavoidable error on that target, and we measure this error before judging the model.

2 Theoretical results

Target leakage (exact identities). Two of the five targets are closed-form functions of supplied inputs:

$$|Z(G)| = \sum_x \mathbf{1}_{x \in Z(G)}, \qquad k(G) = \mathrm{mean}_x \, |C_G(x)| \quad \text{(Burnside)}.$$

The second has the same form as the network's mean-pooling readout. Both identities are verified to floating-point precision on 13 hand-constructed groups spanning $p = 2$ and $p = 3$ (`tests/test_group_theory.py`). Consequently we report a closed-form baseline, which needs no training, next to the model's score for these targets. The model is judged against this baseline, and a low absolute error is not enough by itself.

Persistent homology on the full graph. The identity commutes with everything, so it is a universal vertex and all pairwise distances are 1 or 2. The Vietoris–Rips complex of $\Gamma(G)$ is discrete below the edge scale, a cone over the identity from the edge scale up to distance 2, and a full simplex from distance 2 on, so the complex is contractible at every scale from the edge scale on. Hence $H_1 \equiv 0$, and the dimension-1 persistence descriptor used in this study is trivial for every full commuting graph. We therefore compute persistent homology (PH) on the noncentral

graph (centre removed), the minimal modification under which the descriptor can carry signal, with the persistence-image transform fitted on training diagrams only.

Collision theory and hard error floors. Commuting is invariant under central translation, so $\Gamma(G)$ factors through $G/Z(G)$. A known theorem (Cameron–Kuzma, arXiv:2012.03789; Arvind–Cameron, arXiv:2206.01059) says that isoclinic groups of the same order have isomorphic commuting graphs. The following results build on it.

Proposition 1. *The two non-abelian groups of order p^3 share the graph $K_p \vee ((p{+}1) \cdot K_{p^2-p})$ for every prime p. Hence neither the exponent nor the property of being powerful is recoverable from $\Gamma(G)$: at order 27 one group of the colliding pair has exponent 3 and is not powerful, the other has exponent 9 and is powerful.*

Corollary 2. *In the hierarchy R_0 (bare graph) $\subset R_1$ (+ element orders) $\subset R_2$ (+ power depth), the exponent has recoverability threshold exactly 1.*

Corollary 3 (Collisions and the five targets)**.** *Nilpotency class, derived length, and $|[G,G]|$ are isoclinism-invariant, and $|Z(G)|$ and $k(G)$ are graph-determined. Hence an isoclinic collision can never separate any of the five targets. A damaging collision must be non-isoclinic, and one exists inside the dataset. Arvind and Cameron's order-128 example (Schur covers of $\mathrm{SmallGroup}(64,182)$ vs. $\mathrm{SmallGroup}(64,182) \times C_2$) has isomorphic commuting graphs with commutator sizes differing by a factor of 2. Commutator size therefore carries a strictly positive MSE floor for every model reading the unlabelled graph (R_0). The census refines each collision class by labelled isomorphism and reports floors at R_1 (+ element orders) and R_2 (+ power depth) as well, since a collision need not survive enrichment. For example, on the 13 hand-constructed test groups the exponent's floor is 4.31 at R_0 and 0 at R_1. For the two cases the theory leaves open, the full-dataset census shows that, up to order 128, nilpotency class is separated by non-isoclinic collisions (Bayes floor 0.0005 even at R_2), while derived length is separated by none. The class-2 conjecture of earlier versions of the Arvind–Cameron line was refuted by O'Brien's order-64 examples (J. Algebra 703 (2026)), which the census recovers as a validation case.*

Proposition 4 (Bayes floor)**.** *For any predictor f factoring through a representation R, $\mathbb{E}[(I - f \circ R)^2] \geq \mathbb{E}[\mathrm{Var}(I \mid R)]$.*

The exhaustive collision census computes this floor numerically per target at R_0, R_1, and R_2, and checks that the known order-128 collision is recovered with its factor-2 commutator signature. Candidate collisions are grouped by WL hash, confirmed by exact BLISS canonical forms, and refined by labelled isomorphism using coloured BLISS search (`src/collision_census.py`).

3 Computational verification

The test suite reimplements the group-theoretic computations from multiplication tables, without Sage, and checks them against 13 known groups in 58 tests, all of which pass. The tests include the power-depth values of C_8, D_8, and $\mathbb{Z}_9 \rtimes \mathbb{Z}_3$, and structure-level isomorphism proofs for four collision pairs (C_4/C_2^2, D_8/Q_8, the order-27 pair, D_{16}/SD_{16}). On these 13 groups the census finds six collision classes, zero floor for all five targets, a positive floor for the exponent (4.31), and a 0–1 classification floor of 1/13 for powerfulness. These are the values the theory predicts at those orders. A worked odd-prime example ($\mathbb{Z}_9 \rtimes \mathbb{Z}_3$: powerful but not powerfully nilpotent, proved via the Traustason–Williams rank-2 criterion and confirmed by exhaustive computation) serves as a check on the computation of powerful class at odd primes.

The full-dataset census. The census was run on all 2668 nontrivial 2-groups of order at most 128 (and on the 594-group $p = 3$ dataset). Both literature validation cases were recovered: the Arvind–Cameron order-128 collision with its factor-2 commutator signature, and the reported O'Brien order-64 pair. 2667/2668 groups share their commuting graph with another group (74 collision classes). On this dataset nilpotency class is separated by non-isoclinic collisions. Its Bayes floors are 0.1408 (R_0), 0.0090 (R_1), and 0.0005 (R_2), so the floor is strictly positive even at R_2, the representation given to the GNN. No collision separates derived length (floor 0 at all levels, up to order 128). The commutator-size floors are 0.0217, 0.0066, and 0.0047 at R_0, R_1, and R_2 respectively. The exponent's floor is 17.60 at R_0 and vanishes at R_1, in line with its recoverability threshold of 1. At $p = 3$ the same pattern holds, and powerfulness remains ambiguous even at R_2 (floor 0.0606, against 0.0045 at $p = 2$).

Training results. Training ran with five seeds (42–46, the R_2+PH configuration, CPU-only, mean 451.5 s and $\leq$ 2.4 GB RSS per run). Pooled test MSE, mean $\pm$ std: nilpotency class 1.73 ± 3.22, derived length 0.020 ± 0.019, centre size 0.45 ± 0.32 and commutator size 1.05 ± 1.61 (both $\log_2$), conjugacy classes 188.5 ± 295.1. The best single seed, selected post hoc on the test set and so indicative of scale only, scored 0.174, 0.012, 0.030, 0.029, and 0.246 on the same five targets. The zero-training closed-form baselines scored MSE 0.000 in every run. Gradient boosting on per-graph summary statistics beats the GNN on every target, replicated across all five seed splits (gradient-boosting mean MSE 0.034, 0.000, 0.011, 0.014, and 0.000), and linear regression recovers $k(G)$ exactly. The graph-only ablation is 12.6% above the split-specific R_0 nilpotency test floor (0.155 vs. 0.13765 on its seed's test set, with a dataset-global floor of 0.141), so it comes close to recovering the collision-class means. All measured errors are consistent with both the dataset-global census floors and the per-seed split-specific floors. For every seed, all 268 test groups have an R_0-isomorphic training counterpart, and 41–48% share their full R_2 representation class with a training group. The pooled numbers therefore measure interpolation within largely seen representations. Order transfer, which holds out entire orders, is the stronger test of generalization, and performance degrades under it. A model trained on orders ≤ 64 and tested zero-shot on order 128 fails for the scale-dependent targets (centre 20.9, commutator 11.0 $\log_2$-MSE, $k(G)$ 2285) and has smaller but still large errors for the bounded targets (nilpotency class 0.74, derived length 0.96).

4 Experimental scope

The protocol was fixed before the training runs began. It used five seeds, reported the mean $\pm$ std of each metric per target, pooled, and of the MSE by group order, and recorded wall-clock time and memory. The model was compared with a constant-mean predictor, the closed-form identities, linear and gradient-boosted models on graph summaries, and GNN ablations using R_0, R_1 and R_2 input. The results are given in Section 3. The main R_2+PH model was evaluated over seeds 42–46. The GNN ablations used seed 42, while the non-GNN baselines were evaluated on all five corresponding splits. In the cross-prime comparison, a GNN trained on 2-groups at R_2 was fine-tuned on the 594-group $p = 3$ dataset and reached test MSE 0.259, 0.001, 0.214, 0.227, and 28.1 on the five targets. In a single-seed comparison it beat from-scratch training at the standard learning rate 10^{-3} (same seed, split, and budget, with test MSE 0.466, 0.000, 0.534, 0.426, and 2390.9) on four of the five targets, most markedly on $k(G)$ (28 against 2391), although a single seed cannot establish a general fine-tuning advantage. Training from scratch at the low fine-tuning learning rate diverged. The powerful-groups prediction experiment used a two-head design,

with classification of powerfulness and masked regression of powerful class, and five seeds at each prime. The classification results are the main outcome, and the masked regression estimates are exploratory because the eligible test subsets are small (the full paper gives the table in its section on powerful and powerfully nilpotent p-groups). The study does not include an ablation with the noncentral graph as input, a commutator-weighted representation, or coclass-tree features.

5 Limitations

The commuting graph encodes a binary relation only. Graded non-commutativity (commutator magnitudes on a non-commuting graph, with an edge-aware GNN) was not evaluated in this study. The class-2 conjecture would have limited where further damaging collisions could occur, but O'Brien's order-64 examples refute it. Which targets the remaining collisions separate is therefore known only from the census, which covers orders up to 128 at $p = 2$ and up to 729 at $p = 3$. The main empirical limitation of the five-seed experiment is unstable optimization. With patience-20 early stopping, per-seed results vary widely (one of five seeds is an outlier on nilpotency class and $k(G)$), and the GNN's five-seed mean does not beat gradient boosting on per-graph summary statistics on any target. The Bayes floors are lower bounds for every model that reads these representations, and the pooled errors of the R_2+PH model lie well above them.